

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (Electrical Engineering)

December 2011

EE 304 Industrial Instrumentation

Date: 13.12.2011, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a)** Explain how McLeod gauge measure the vacuum pressure. [04]
- (b)** Explain the following terms: [03]
- (i) Absolute pressure
 - (ii) Gauge pressure
 - (iii) Atmospheric pressure
- Q - 2 (a)** Explain different mechanical devices used for measurement of pressure in detail. [08]
- (b)** Define followings: [06]
- [1] Resolution [2] Sensitivity [3] Accuracy [4] Precision [5] Reproducibility
- [6] Drift

OR

- Q - 2 (a)** Discuss in detail: the types and characteristics of thermocouples for temperature measurement. [07]
- (b)** Describe the construction and working of disappearing filament optical pyrometers. Also gives its advantages and disadvantages. [07]
- Q - 3** Explain any two: [14]
- [1] Temperature measurement using RTD
 - [2] Capacitive methods for liquid level measurement
 - [3] Inductive Transducers for pressure measurement
 - [4] Resistive hygrometer

SECTION - II

- Q - 4 (a)** Describe the methods used for measurement of Linear velocity and explain any one in detail. [05]
- (b)** Explain the difference between measurement and instrumentation. [02]
- Q - 5 (a)** What is Piezo-resistive effect? Define gauge factor. Derive an expression for the gauge factor for strain gauge in terms of Poisson's ratio. [07]

- (b) A strain gauge having a resistance of 100Ω and gauge factor of 2 is connected in series with a ballast resistance of 100Ω across a 12 volt supply. Calculate difference between output voltage when [1] no stress is applied and [2] stress of 140 MN/m^2 is applied. Take modulus of elasticity is 200 GN/m^2 . [03]

- (c) Why are dummy gauges used? In what way do they affect the output of a strain gauge bridge? [04]

OR

- Q - 5 (a) Why temperature compensation is required in strain gauge measurement? Derive the equation of output voltage when two active gauges are connected in two adjacent arms of Wheatstone bridge to compensate the temperature effect. [06]

- (b) Four strain gauges are mounted on a simple flat tensile specimen arranged for a complete temperature compensation and maximum sensitivity when connected in a four arm bridge circuit. An $800 \text{ K}\Omega$ calibration resistor is shunted across one of the strain gauges. Gauge resistances are each 188Ω and gauge factors are 1.22. [03]

If strain gauge readout is 150 divisions when the calibration switch is closed and 240 divisions when the load is applied, what is the axial strain in the specimen?

- (c) Explain load cells in detail. [05]

Q - 6 Attempt any two:

- [1] Ultrasonic flow transducer
[2] Electromagnetic flow meter
[3] X-Y recorders
[4] Magnetic tape recorders [14]
